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### Combined Lock and Tension Connector for Support Rods

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#### Abstract

A connector for telescoping rods is provided including a core shaft and a locking sleeve. The core shaft includes a threaded portion having a first threading and a conical portion having a first frustoconical surface. The locking sleeve can receive at least a portion of the core shaft and defines a second threading that engages the first threading. In use, the internal locking connector is coupled to an inner rod and an outer rod such that rotating the inner rod and the outer rod relative to one another both increases tension of the rods against opposing mounting surfaces and locks the outer rod in a position relative to the inner rod.

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#### Background/Summary

## FIELD

[0001] This disclosure relates to support rods, and, more particularly, to lock and tensioning connectors for support rods.

## BACKGROUND

[0002] Support rods are used horizontally to support shower curtains, drapery, towels, clothes, or other items. They also are used vertically to hold baskets and other items. One common vertical use is a shower caddy. Typical support rods are adjustable and include an outer rod and an inner rod that slides telescopically in and out of the outer rod to adjust the overall length of the rods. A connector locks the outer and inner rods together at the desired adjusted length.

[0003] There is a need to improve connectors so that sufficient locking force is applied to prevent slippage between the two rods. There is also a need to simplify construction and use of a support rod so that a single mechanism both tensions the rods against the support surfaces and locks the two rods at the desired adjusted length.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is an exploded view of a rod assembly according to an embodiment.

[0005] FIG. 2 is a front-side, first end perspective view of a connector of the rod assembly of FIG. 1.

[0006] FIG. 3 is a back-side, second end perspective view of the connector of FIG. 2.

[0007] FIG. 4 is a centrally taken cross-sectional view of the connector of FIG. 2 taken along line 8-8 of FIG. 3, shown inserted into an inner rod and an outer rod.

[0008] FIG. 5 is a front-side, second end perspective view of an insert of the connector of FIG. 2.

[0009] FIG. 6 is back-side, first end perspective view of the insert of the connector of FIG. 2.

[0010] FIG. 7 is a front-side, first end perspective view of a sleeve of the connector of FIG. 2.

[0011] FIG. 8 is a back-side, second end perspective view of the sleeve of the connector of FIG. 2.

### DETAILED DESCRIPTION

[0012] With reference to FIG. 1, there is illustrated a rod assembly **100** with a hollow, inner rod **102** and a hollow, outer rod **104**. The inner rod **102** is telescopically received in the outer rod **104** to adjust the overall length of the rod assembly **100**. Two end caps **106**, **108** may be fitted to the inner rod **102** and the outer rod **104**, respectively, to engage opposing support surfaces. The end caps may be adjustable or non-adjustable end caps, such as those described in U.S. Pat. Nos. 10,959,559, 11,382,447, U.S. Publication No. 2023/0277012, and in U.S. patent application Ser. No. 18/389,502, filed on Nov. 14, 2023, all of which are incorporated by reference herein in their entireties.

[0013] A connector **110** engages both the inner rod **102** and the outer rod **104** along a

[0014] central longitudinal axis *y* of the rod assembly **100**. The connector **110** is a single mechanism having a dual purpose. Specifically, adjusting the connector **110** both locks the inner rod **102** and the outer rod **104** together and increases the tension of the inner rod **102** and the outer rod **104** against the opposing support surfaces.

[0015] More specifically, the connector **110** is configured so that simply rotating the inner rod **102** and the outer rod **104** relative to one another both locks the rods **102**, **104** together at the desired length and moves the rods **102**, **104** away from one another towards the opposing support surfaces to increase tension against the support surfaces.

[0016] With reference to FIGS. 1-4, the connector **110** includes an insert **120** and a sleeve **160**. A first end **124** of the insert **120** is configured to engage the inner rod **102**, and a second end **126** of the insert **120** is configured to be inserted into the outer rod **104**. The sleeve **160** has a split ring configuration and, in use, is disposed about the insert **120**. The sleeve **160** is configured to displace

relative to the insert **120** along a longitudinal axis L of the connector **110**, as described further below, to selectively lock and unlock the inner rod **102** and outer rod **104**.

[0017] With reference to FIGS. **1**, **4** and **5-6**, the insert **120** is an elongated one-piece body **122**. The body **122** includes an insert portion **128** extending from the first end **124**, a conical or frustoconical portion of the body extending from the second end **126**, and a shaft portion **140** extending therebetween that is partially threaded.

[0018] The insert portion **128** is generally cylindrical and is sized to be inserted within the inner rod **102**. In embodiments, the insert portion **128** is sized to have a friction fit connection with the inner surface of the inner rod **102** so that the insert **120** is fixed from rotation relative to the inner rod **102**. In some approaches, an outer side surface **129** of the insert portion **128** engages the inner surface of the inner rod **102**. In some approaches, for instance as illustrated, a plurality of ribs (e.g., tapered ribs) may extend longitudinally along the outer side surface **129** to enhance the friction fit connection. In certain approaches, the insert portion **128** is inserted into the inner rod **102** during a manufacturing step and dimpled to further enhance friction and ensure the insert **120** does not rotate within the rod.

[0019] The insert portion **128** may also include a chamfered end portion **132** that facilitates insertion of the insert portion **128** into the inner rod **102**. In embodiments, the insert portion **128** is hollow. For instance, the insert portion **128** may be defined by an annular wall **127** defining a central opening into a cavity **135**. In some approaches, the insert portion **128** is hollow but the other portions of the insert **120** (e.g., the shaft portion **140** and the conical portion **150**) are solid. In other approaches, the insert portion **128** is solid. By another approach, a central passage extends through the entire insert **120**.

[0020] The insert **120** further includes an annular flange **138** that extends radially outwardly between and contiguous with the insert portion **128** and the shaft portion **140**, having a greater radial extent than the insert portion **128** and the shaft portion **140**. The annular flange **138** is radially positioned so that an end of the inner rod **102** may abut the annular flange **138** when the insert portion **128** is inserted therein. Thus, the annular flange **138** acts as a stop to prevent further insertion of the insert **120** into the inner rod **102**. Specifically, the stop prevents the shaft portion **140** and the conical portion **150** from being inserted into the inner rod **102**. As discussed below, the annular flange **138** is also positioned to stop displacement of the sleeve **160** relative to the insert **120**.

[0021] The outer diameter of the annular flange **138**, in embodiments, is less than the inner diameter of the outer rod **104**. For instance, as shown in FIG. **4**, when the connector lock **110** is inserted within the inner rod **102** and the outer rod **104**, the flange **138** and the outer rod **104** may define an annular space or gap **129a** therebetween so that the flange **138** does not contact the outer rod **104**. This ensures sufficient clearance for rotational displacement to occur between the inner rod **102** and the outer rod **104**.

[0022] In some approaches, instead of a continuous annular flange **138** as illustrated, the flange **138** may be a discontinuous annular flange or may include a plurality of flanges spaced about the insert **120** to prevent further insertion of the insert **120** into the inner rod **102**.

[0023] The shaft portion **140** is generally cylindrical and extends between the flange **138** and the conical portion **150**. The shaft portion **140** includes external threading **142** disposed along a portion of the shaft portion. As illustrated, the external threading **142** does not extend to the flange **138** and the conical portion **150**. The external threading engages internal threading **169** of the sleeve **160**. The threading may be partial, discontinuous threading or may be continuous threading.

[0024] In embodiments, the diameter of the shaft portion **140** is less than the diameter of the insert portion **128**. The narrowness of the shaft portion **140** allows a space for the locking sleeve **160** to extend about the shaft portion **140** while fitting within the outer rod **104**.

[0025] The conical portion **150** extends from the shaft portion **140**. The conical portion **150** has a gradually increasing diameter from a first end **154** of the conical portion **150** adjacent the shaft

portion **140** to a second end **156**. The gradually increasing diameter defines a frustoconical outer surface **152** of the conical portion **150** that acts as a wedge during the locking processes. In embodiments, the outer surface **152** is substantially angular. In certain approaches, the slope of the frustoconical outer surface **152** is such that an angle  $\alpha$  (FIG. **4**) defined between the frustoconical outer surface **152** and the surface at the second end **156** is at least about 45 degrees and at most about 89 degrees. In some embodiments, the angle  $\alpha$  is at least about 60 degrees and at most about 85 degrees.

[0026] The maximum diameter of the conical portion **150** at the second end **156**, in embodiments, is greater than the diameter at any other portion of the insert **120**. For instance, in one approach, the conical portion **150** has a maximum diameter sized so that there is slight contact between the frustoconical outer surface **152** at the second end **156** with the inner surface of the outer rod **104**. In some approaches, the frustoconical outer surface **152** does not contact the outer rod **104**, and there is a slight gap therebetween. In any case, the maximum diameter of the conical portion **150** is selected relative to the inner diameter of the outer rod **104** to enable the conical portion **150** to force the sleeve **160** outwardly against the outer rod **104** to increase the tension between the sleeve **160** and the outer rod **104**, as described further below.

[0027] The insert **120** may be formed from polymeric materials such as polycarbonate, polystyrene, polypropylene, acrylonitrile butadiene styrene (ABS), styrene acrylonitrile (SAN), polyurethane, polyvinyl chloride (PVC), nylon, rubber, etc. In some approaches, the material of the insert **120** may be a metal such as aluminum or stainless steel. The size of the insert **120** may vary based on the sizes of the rods the insert **120** is intended for. In one approach, the insert may have a main diameter at the insert portion **128** (not including the ribs **130**) of about 0.625 inches (15.875 mm) to about 1.25 inches (31.75 mm) and a length of about 0.5 inches (12.7 mm) to about 2 inches (50.8 mm). In some approaches, the length of the insert portion **128** is about one fourth to about one half of the length of the entire insert **120**, such as about one third.

[0028] With reference to FIGS. **4** and **7-8**, the sleeve **160** has a generally cylindrical, split ring configuration. Specifically, a cylindrical wall **162** defines a central passage **161** of the sleeve **160** which is sized to receive at least a portion of the insert **120** (e.g., the shaft portion **140**). The split ring structure of the sleeve **160** defines a gap **164** in the cylindrical wall **162** which extends longitudinally an entire length of the cylindrical wall **162**. The gap **164** provides flexibility to the sleeve **160** so that it can expand outwardly during the locking process. In addition, a coring **180** is defined in an exterior surface of the cylindrical wall **162** substantially opposite the gap **164**, providing a thinned portion of the cylindrical wall **162**. The coring **180** may extend longitudinally an entire length of the cylindrical wall and provides further flexibility to the sleeve **160**. For instance, the coring, in embodiments, serves as a “living hinge” to facilitate outward expansion of the portions of the cylindrical wall **162** on opposite sides of the coring **180** during the locking process.

[0029] In embodiments, an angled projection **182** projects from the cylindrical wall **162** at one side of the coring **180**. In embodiments, the angled projection **182** is positioned to engage the outer rod **104** when the sleeve **160** is inserted therein to create friction between the sleeve **160** and the outer rod **104** to resist rotation of the sleeve **160** during the locking processes. In some embodiments, the angled projection **182** functions to provide sufficient engagement between the sleeve **160** and the outer rod **104** to ensure the sleeve **160** will rotate with the outer rod **104** when the rods **102**, **104** are twisted relative to one another during the locking function, while at the same time not creating too much friction to hinder telescoping of the rods **102**, **104** relative to one another.

[0030] The sleeve **160** is sized so that it can be inserted into the outer rod **104**. Specifically, an outer diameter of the sleeve **160** may be sized to permit light frictional engagement of an exterior cylindrical side surface **163** of the sleeve **160** with an inner surface of the outer rod **104** upon insertion of the sleeve **160** into the outer rod **104**. Such light friction connection between the sleeve **160** and the outer rod **104** upon insertion may be sufficient to fix the sleeve **160** from rotation

relative to the outer rod **104** during the locking process but also permit telescoping of the outer rod **104** relative to the inner rod **104**. In some embodiments, the outer diameter of the sleeve **160** may be sized so that the sleeve **160** does not engage the outer rod **104** upon insertion into the outer rod **104** (e.g., there may be a slight gap), and the angled projection **182** is employed to rotationally fix the sleeve **160** relative to the outer rod **104**. In some approaches, the exterior side surface **163** is entirely smooth to maximize contact with the inner surface of the outer rod **104**. In other approaches, the exterior side surface **163** may include ribs or other projections to achieve a desired amount of friction.

[0031] An interior cylindrical surface **175** of the sleeve **160** includes internal threading **169** configured to cooperate with external threading **142** of the insert **120**. A threaded portion **175** of the interior surface **175** containing the internal threading **169**, in embodiments, may extend from a first end **165** of the sleeve **160** and along at least a portion of the length of the sleeve **160** towards a second end **167** of the sleeve **160**. As illustrated, the internal threading **169** does not extend all the way to the second end **167**. The threading may be partial, discontinuous threading or may be continuous threading.

[0032] The interior surface **175** also includes a smooth (non-threaded) angled or sloped portion **172** extending between the internal threading **169** and the second end **167**. The interior angled surface **172** is defined by an annular tapered portion of the cylindrical wall **162**. In embodiments, the tapering of the cylindrical wall **162** starts at the end of the internal threading **169** closest to the second end **167** and gradually tapers towards the second end **167**, so that a thickness of the cylindrical wall **162** at the second end **167** is less than a thickness of the cylindrical wall at the internal threading **169**. Specifically, the thickness of the cylindrical wall **162** at the first end **165** defines an annular face **166** at the first end **165** that may seat against or abut the flange **138** of the lock insert **120** when the shaft portion **140** is fully threaded into the locking sleeve **160**.

[0033] In embodiments, the angle or slope of the interior angled surface **172** is selected so that when the interior angled surface **172** moves along the frustoconical surface **152** of the insert **120** during the locking process (described further below), engagement between the two surfaces **172**, **152** causes the frustoconical surface **152** to exert pressure on the interior angled surface **172** and force the sleeve **160** to expand outwardly against the outer rod **104**. In some configurations, the slope of the frustoconical surface **152** is greater than the slope of the interior angled surface **172**, at least when the interior angled surface **172** is in its neutral position inserted into the outer rod **104** and not engaging the frustoconical surface **152**.

[0034] In some approaches, an angle  $\beta$  (FIG. 4) defined between the interior angled surface **172** and a line extending from an end surface at the second end **167** of the sleeve **160** is at least about 45 degrees. A maximum may, for example, be about 89 degrees. In some embodiments, the angle  $\beta$  may be at least about 60 degrees or at least about 70 degrees.

[0035] In certain configurations, the angle  $\beta$  may be about 90 degrees and thus there is no tapered portion or angled surface. In some approaches, the angle  $\beta$  may be up to 95 degrees.

[0036] The sleeve **160** may be formed from a polymeric material such as polycarbonate, polystyrene, polypropylene, ABS, SAN, polyurethane, PVC, rubber, etc.

[0037] The size of the sleeve **160** may vary for use with rods of different sizes. In one example, the sleeve may be about 0.787 inches (20 mm) to about 1.181 inches (30 mm) in length with a length of the threaded portion being about 0.393 inches (10 mm) to about 0.787 inches (20 mm) and a length of the tapered portion being about 0.195 inches (5 mm) to about 0.590 inches (15 mm).

[0038] In some approaches, a thickness of the wall of the sleeve **160** at the threaded portion may be about 0.1 inches to about 0.2 inches, for example about 0.118 inches while the maximum thickness of the tapered portion may be about 0.1 inches to about 0.2 inches, for example about 0.118 inches and the minimum thickness about 0.02 to about 0.05 inches, for example about 0.039 inches. The thickness of the sleeve **160** at the cored out portion may, for example, be about 0.040 to about 0.060 inches, for instance about 0.049 inches, or at most about half the thickness of the sleeve **160**

at the threaded portion.

[0039] FIG. 4 illustrates an initial, unlocked position of the connector **110** when the connector **110** is installed in the inner rod **102** and the outer rod **104**. In this position, the connector **110** is selectively positioned relative to the outer rod **104** to adjust the total length of the rod assembly **100**. During installation, the insert portion **128** of the rod connector **110** may already be inserted into the inner rod **102** so that the remainder of the insert **120** and the sleeve **160** coupled thereto protrude from the inner rod **102**. For example, this pre-assembly may have occurred during a manufacturing step, which may include dimpling to adhere the components firmly together. In other approaches, the components **128**, **102** may be glued or welded. In certain approaches, the rod connector **110** may not already be inserted into the inner rod **102** and the user inserts the insert portion **128** of the rod connector **110** into the inner rod **102**.

[0040] During installation, the user adjusts the total length of the rod assembly **100** so that the rod assembly can extend between a first mounting surface and a second mounting surface (e.g., two opposing walls). To do so, the inner rod **102** and protruding parts of the connector **110** are inserted into the outer rod **104** and telescoped until the rod assembly **100** is the correct length. The farther the inner rod **102** and connector **110** are adjusted into the outer rod **104**, the smaller the length of the rod assembly **100**.

[0041] In the initial, unlocked position, the insert portion **128** of the insert **120** is inserted into end of the inner rod **102** with a connection that fixes the inner rod **102** and the insert **120** from rotation relative to one another when the inner rod **102** is rotated. The end of the inner rod **102** may abut the flange **138** of the insert **120** so that the rest of the insert **120** is not inserted into the inner rod **102** and instead protrudes therefrom.

[0042] The sleeve **160** is arranged about the shaft portion **140** of the insert **120** in the initial, unlocked position, fully threaded onto the shaft portion **140** of the insert **120** until the sleeve **160** abuts the flange **138**. In some approaches, in the unlocked position, substantially the entirety of the sleeve **160** is arranged about or coextensive with the shaft portion **140**. In this position, the interior angled surface **172** of the sleeve **160** does not engage the frustoconical surface **152** of the insert **120**. In embodiments, in the initial unlocked position, the entirety of the conical portion **150** protrudes from the sleeve **160**.

[0043] In some configurations, the sleeve **160** may be dimensioned so that it is always captivated on the insert **120** during normal use of the connector **110**. For instance, a minimum inner diameter of the sleeve **160** defined by the internal threading **169**, and/or a maximum flexibility of the locking sleeve **160**, may limit the sleeve **160** from being completely removed from the insert **120** so that it is always between the flange **138** and the second end **156** of the conical portion **150**. In other configurations, the sleeve **160** may be flexible enough to be removed from the insert **120**.

[0044] In the unlocked position, the exterior surface **163** and/or the angled projection **182** of the sleeve **160** engage the inner surface of the outer rod **104**. The amount of friction between the sleeve **160** and the outer rod **104** may be selected so that the outer rod **104** and the sleeve **160** are fixed from rotation relative to one another when the outer rod **104** is rotated, though still permitting the sleeve **160** and the outer rod **104** to slide longitudinally relative to one another with sufficient ease to adjust the length of the rod assembly **100**.

[0045] As illustrated, when installed, the flange **138** of the insert **120** does not contact the outer rod **104**, though the widest portion of the conical portion **150** may, in some embodiments, contact the outer rod **104**.

[0046] After the length of the rod assembly **100** is adjusted so that the rods **102**, **104** engage the opposing mounting surfaces, further tension is needed for a strong connection. In addition, further pressure is needed between the sleeve **160** and the outer rod **104** so that the outer rod **104** does not slip or move relative to the inner rod **102** during use of the rod assembly **100**. Adjusting the connector **110** achieves both locking and tensioning.

[0047] Specifically, the user adjusts the connector **110** by rotating the inner rod **102** and the outer

rod **104** relative to one another. For instance, in embodiments, the user twists both the inner rod **102** and the outer rod **104** simultaneously in opposite directions. The user may also firmly grasp one of the rods and simply rotate the other. When the inner rod **102** and the outer rod **104** are rotated relative to one another, the threaded connection between the sleeve **160** and the insert **120** causes the sleeve **160** and the insert **120** to move relative to one another. Specifically, the sleeve **160** is moved away from the flange **138** and towards the conical portion **150** of the insert **120**. [0048] When the sleeve **160** is moved towards the conical portion **150** (and/or the conical portion **150** is moved towards the sleeve **160**), the interior angled surface **172** of the sleeve **160** engages the frustoconical surface **152** of the conical portion **150**. The frustoconical surface **152** wedges against the angled surface **172** of the sleeve **160**, exerting pressure against the angled surface **172** as the tapered end portion **170** of the sleeve **160** is forced into the gradually decreasing space between the frustoconical surface **152** and the outer rod **104**. Specifically, the engagement between the surfaces results in the radially outward expansion of the locking sleeve **160** against the inner surface of the outer rod **104**. The flexibility of the split-ring sleeve **160** facilitates the expansion. The outward expansion of the sleeve **160** against the outer rod **104** increases the friction force between the outer rod **104** and the sleeve **160** so that the outer rod **104** is firmly locked to the sleeve **160** (e.g., inhibited from linear or axial movement relative thereto) and, thus, locked in position with respect to the inner rod **102** without any slippage during use of the rod assembly **100**. In addition, as the sleeve **160** becomes locked to the outer rod **104** and axially fixed relative to the outer rod **104**, final twisting of the rods **102**, **104** causes the sleeve **160** and the outer rod **104** to move together away from the inner rod **102**, slightly increasing the total length of the rod assembly **100** and forcing the rods **102**, **104** in greater tension against the opposing surfaces. Thus, a final “fine” adjustment of the rod assembly **100** against the mounting surfaces that increases the mounting tension is provided.

[0049] Accordingly, the connector **110** provides a simple mechanism for both increasing the tension of the rod assembly **100** against the mounting surfaces and locking the outer rod **104** and the inner rod **102** in position relative to one another.

[0050] To ensure that the rod assembly **100** is fully locked with maximum tension, the rods **102**, **104** may be rotated relative to one another until rotation can no longer occur. This occurs when the angled surface **172** of the locking sleeve **160** and the frustoconical surface **152** of the lock insert **120** fully abut, the conical portion **150** being drawn almost entirely into the locking sleeve **160**, and the distal end of the conical surface **172** applying radial pressure to the distal end of the angled surface **172** of the sleeve **160**. That is, the final position of the rod assembly **100** achieved when the rods **102**, **104** cannot be rotated any further, provides both maximum radial pressure to lock to rods **102**, **104** in position and maximum tension of the rod assembly against the mounting surfaces.

[0051] To uninstall or unlock the rod assembly **100**, the user reverses the twisting operation by twisting each of the outer rod **104** and the inner rod **102** in a direction opposite of the direction required for locking. This causes the outer rod **104** and the inner rod **102** to move towards each other via the threaded connection between the sleeve **160** and the insert **120**. This decreases the tension against the mounting surfaces and/or disengages the rod assembly **100** from the mounting surfaces. In addition, this movement results in the interior angled surface **172** of the sleeve **160** disengaging from the frustoconical surface **152** of the conical portion **150** so that the sleeve **160** returns to its initial, non-expanded state. As a result, the outer rod **104** is slidable relative to the inner rod **102** and connector **110** so that the rods **102**, **104** can be taken apart or so that the length of the rod assembly **100** can be further adjusted.

[0052] As noted above, a benefit of the connector **110** is that it provides a single mechanism that both increases the mounting tension of the rod assembly **100** and locks the rods **102**, **104** in position relative to one another without slippage. Because the connector **110** permits a final fine adjustment of the rods **102**, **104** against the mounting surfaces to increase the tension, the rod assembly **100** does not require other tensioning mechanisms, such as adjustable end caps.

[0053] A further benefit of the connector **110** is that it is simple to manufacture as it only includes two components (the insert **120** and the sleeve **160**). Another advantage is that the connector **110** is contained entirely within the rods **102**, **104** and thus obscured from view to maintain a clean appearance of the rod assembly **100**.

[0054] In certain embodiments, a method of mounting an adjustable rod between two opposing mounting surfaces may include providing an inner rod with a connector coupled to one end, the connector including a core shaft and a sleeve disposed about the core shaft, the core shaft including first threading and a conical portion and the sleeve including second threading cooperating with the first threading to move the sleeve and the core shaft relative to one another. The method may further include telescoping the inner rod and the connector within an outer rod to adjust a total combined length of the adjustable rod and rotating the inner rod and the outer rod relative to one another, wherein rotating the inner rod and the outer rod relative to one another moves the sleeve into engagement with the conical portion to expand the sleeve radially outwardly against the outer rod to lock the outer rod in position relative to the inner rod, and causes the inner rod and the outer rod to translate in opposite directions to increase a tension of the adjustable rod against the two opposing mounting surfaces.

[0055] While there have been illustrated and described particular embodiments of the present invention, those skilled in the art will recognize that a wide variety of modifications, alterations, and combinations can be made with respect to the above-described embodiments without departing from the scope of the invention, and that such modifications, alterations, and combinations are to be viewed as being within the ambit of the inventive concept.

## Claims

1. A connector for telescoping rods comprising: a core including a shaft portion with a first threading and a frustoconical portion having a first frustoconical surface; and a sleeve capable of receiving at least a portion of the core and having a second threading and a second frustoconical surface, wherein the second threading is engageable with the first threading to selectively displace the sleeve relative to the core between a first position where the first frustoconical surface and the second frustoconical surface are separated and a second position where the first frustoconical surface and the second frustoconical surface engage.
2. The connector of claim 1, the sleeve being radially expanded in the second position due to the engagement between the first frustoconical surface and the second frustoconical surface.
3. The connector of claim 1, the sleeve including an annular wall defining a gap extending along an entire length of the sleeve.
4. The connector of claim 3, wherein the sleeve defines a coring in the annular wall substantially opposite the gap to provide flexibility to the sleeve.
5. The connector of claim 1, wherein the sleeve includes an annular wall, the annular wall having a tapered portion forming the second frustoconical surface.
6. The connector of claim 5, wherein the tapered portion is disposed between the second threading and a terminal end of the sleeve.
7. The connector of claim 1, wherein the first threading of the core is intermediate the frustoconical portion and a non-threaded insert portion of the core.
8. The connector of claim 7, wherein the non-threaded insert portion has a first maximum diameter, the shaft portion has a second maximum diameter at the first threading, and the first maximum diameter is greater than the second maximum diameter.
9. The connector of claim 8, wherein the frustoconical portion has a third maximum diameter that is greater than the first maximum diameter of the non-threaded insert portion.
10. The connector of claim 7, further comprising a radial flange positioned between the non-threaded insert portion and the shaft portion.



- 11.** The connector of claim 1, wherein the frustoconical portion increases in radial dimension from an inboard end to an outboard end of the frustoconical portion.
  - 12.** The connector of claim 1, wherein the shaft portion includes a first smooth surface between the first threading and the frustoconical portion.
  - 13.** The connector of claim 12 wherein the core includes a non-threaded insert portion, and the shaft portion includes a second smooth surface between the first threading and the non-threaded insert portion.
  - 14.** An adjustable rod comprising: an inner rod and an outer rod mountable between opposing mounting surfaces, the inner rod capable of telescopically moving into and out from the outer rod to adjust a combined total length between the opposing mounting surfaces; a core couplable at least in part to the inner rod and including a first threading and a wedge; and a sleeve including a second threading that cooperates with the first threading, the sleeve disposable about the core and within the outer rod and sized to engage the outer rod, wherein rotating the inner rod relative to the outer rod in a first direction increases a tension of the adjustable rod against the opposing mounting surfaces and increases pressure between the sleeve and the outer rod to lock the inner rod and the outer rod in position relative to one another.
  - 15.** The adjustable rod of claim 14, the core including an insert portion insertable within the inner rod to frictionally engage the inner rod to restrict rotation of the core relative to the inner rod.
  - 16.** The adjustable rod of claim 15, the core including a shaft portion disposed intermediate the insert portion and the wedge, the shaft portion including the first threading.
  - 17.** The adjustable rod of claim 14, wherein the sleeve is sized so that at least a portion of the sleeve frictionally engages the outer rod to restrict rotation of the sleeve relative to the outer rod while permitting axial movement relative to the outer rod.
  - 18.** The adjustable rod of claim 14, the sleeve including an inner frustoconical surface capable of engaging the wedge when the inner rod is rotated relative to the outer rod in the first direction, the wedge causing expansion of the sleeve to increase the pressure between the sleeve and the outer rod to lock the inner rod and the outer rod in position relative to one another.
  - 19.** The adjustable rod of claim 18, the sleeve including an annular wall, the annular wall including the inner frustoconical surface.
  - 20.** The adjustable rod of claim 14, the sleeve defining an annular wall having a split therein extending longitudinally an entire length of the sleeve and a coring substantially opposite the split to provide flexibility to the sleeve.
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